

Course Code	FINANCIAL MANAGEMENT	L	T	P	C
CB4404		3	0	0	3

COURSE OBJECTIVES:

1. Understand the fundamental concepts of financial management
2. Understand valuation of securities.
3. Analyse operating and financial leverages.
4. Comprehend and apply the concepts of capital budgeting.
5. Understand cash management.

UNIT I FOUNDATIONS OF FINANCE

9

Financial management – An overview- Time value of money- Introduction to the concept of risk and return of a single asset and of a portfolio- Valuation of bonds and shares-Option valuation.

UNIT II INVESTMENT DECISIONS

9

Capital Budgeting: Principles and techniques - Nature of capital budgeting- Identifying relevant cash flows - Evaluation Techniques: Payback, Accounting rate of return, Net Present Value, Internal Rate of Return, Profitability Index - Comparison of DCF techniques - Project selection under capital rationing - Inflation and capital budgeting - Concept and measurement of cost of capital - Specific cost and overall cost of capital.

UNIT III FINANCING AND DIVIDEND DECISION

9

Financial and operating leverage - capital structure - Cost of capital and valuation – designing capital structure. Dividend policy - Aspects of dividend policy - practical consideration - forms of dividend policy - forms of dividends - share splits.

UNIT IV WORKING CAPITAL MANAGEMENT

9

Principles of working capital: Concepts, Needs, Determinants, issues and estimation of working capital - Accounts Receivables Management and factoring - Inventory management – Cash management - Working capital finance: Trade credit, Bank finance and Commercial paper.

UNIT V LONG TERM SOURCES OF FINANCE

9

Indian capital and stock market, New issues market Long term finance: Shares, debentures and term loans, lease, hire purchase, venture capital financing, Private Equity.

NUMBER OF THEORY PERIODS:

45

COURSE OUTCOMES:

On completion of this course, the students will be able to:

- CO1 Understand the fundamental concepts of financial management.
- CO2 Apply valuation of securities and calculate the risk & return in portfolio management.
- CO3 Analyse the cost structure of a company using operating and financial leverages.
- CO4 Develop capital budgets and to estimate working capital.
- CO5 Apply cash management in business.

TEXTBOOKS:

1. M.Y. Khan and P.K.Jain Financial management, Text, Problems and cases Tata McGraw Hill, 6th edition, 2011.
2. M. Pandey Financial Management, Vikas Publishing House Pvt. Ltd., 10th edition, 2012.

REFERENCES:

1. Aswat Damodaran, Corporate Finance Theory and practice, John Wiley & Sons, 2011.
2. James C. Vanhorne –Fundamentals of Financial Management– PHI Learning, 11th Edition, 2012.
3. Brigham, Ehrhardt, Financial Management Theory and Practice, 12th edition, Cengage Learning 2010.
4. Prasanna Chandra, Financial Management, 9th edition, Tata McGraw Hill, 2012.
Srivatsava, Mishra, Financial Management, Oxford University Press, 2011.

STUDY MATERIAL

UNIT I

UNIT – I FOUNDATIONS OF FINANCE:

Financial management – An overview- Time value of money- Introduction to the concept of risk and return of a single asset and of a portfolio- Valuation of bonds and shares-Option valuation.

INTRODUCTION

Financial Management is concerned with planning, directing, monitoring, organizing and controlling monetary resources of an organization. Financial Management simply deals with management of money matters. Management of funds is a critical aspect of financial management. The process of financial management takes place: at the individual as well as organization levels. Our area of dealing is from the view- point of organization.

‘Financial Management’ is a combination of two words, ‘Finance’ and ‘Management’. Finance is the lifeblood of any business enterprise. No business activity can be imagined, without finance. It has been rightly said that business needs money to make more money. However, money begets money, when it is properly managed. Efficient management of business is closely linked with efficient management of its finances. Financial Management is that specialized function of general management, which is related to the procurement of finance and its effective utilization for the achievement of common goal of the organization.

MEANING OF FINANCE

Finance is defined as the provision of money at the time, it is required. Finance is the art and science of managing money. There is no human being, without blood. Similarly, there is no organization that does not require finance, irrespective of the activity, it is engaged in. The way blood is needed for a person to live, so is the requirement of finance to any firm for its survival and growth. Without adequate finance, no organization can possibly achieve its objectives.

Ray G. Jones and Dean Dudley observe that the word ‘finance’ comes directly from the Latin word ‘finis. As a management function, finance has special meaning. Finance function may be defined as the procurement of funds and their effective utilization.

Howard and Upton (1952) defined finance as “the administrative area or set of administrative function in an organization which have to do with the management of flow of cash so that the organization will have the means to carry out its objectives as satisfactory as possible and, at the same time, meet its obligations as they become due.”

Distinction between Money and Finance: Money is expressed in currency. Money can be any country’s currency, which is in the hands of any person or organization. Finance is also money, any country’s currency, which is owned by any person or organization, but lent to others, used

to buy an asset or make investment opportunities. The distinction between money and finance can be explained in another way.

If you hold currency, it is money, while you lend it over to others for buying or investing in investment opportunities, it becomes finance.

It is curious to find that the same currency changes its role from 'Money' to 'Finance', with the change of hands. Let us illustrate. Money raised by a bank, in the form of deposits from public, becomes finance when it is lent to borrowers. If it is granted to buy/ construct a home, it becomes a home loan. It is a vehicle loan, when the amount is lent for buying a car. The amount becomes 'Project Finance', if lent to entrepreneur to start or expand a project.

If you hold money, it does not give any return. You part money in the form of finance, either by way of loan or investment, it starts getting return. Is it not interesting?

MEANING AND DEFINITION—FINANCIAL MANAGEMENT:

The general meaning of finance refers to the provision of funds, as and when needed. However, as management function, the term 'Financial Management' has a distinct meaning. Financial management deals with the study of procuring funds and its effective and judicious utilization, in terms of the overall objectives of the firm, and expectations of the providers of funds. The basic objective is to maximize the value of the firm. The purpose is to achieve maximization of share value to the owners, i.e. equity shareholders.

The objective of every company is to create value for its shareholders. Market price of equity share is the barometer for showing the real 'Value'. The basic objective of financial management is to maximize the shareholders' wealth, represented by the market value of equity shares.

Financial Management is concerned with three activities:

- Anticipating financial needs, which means estimating requirements of the firm in terms of long term and short-term needs or investment in fixed and current assets.
- Acquiring financial resources from different sources to meet the financial needs.
- Allocating funds to maximize shareholders' wealth.

The term financial management has been defined differently by various authors. Some of the authoritative definitions are given below:

1. "Financial Management is concerned with the efficient use of an important Economic resource, namely, Capital Funds." – Solomon
2. "Financial Management is concerned with the managerial decisions that result in the acquisition and financing of short-term and long-term credits for the firm." – Phillioppatus

3. “Financial Management is concerned with the acquisition on , financing and Management of assets with some overall goal in mind.”– James C. Van Horne

4. “Financial Management deals with procurement of funds and their effective utilization in the business.” – S.C. Kuchhal From the above definitions, two basic aspects of financial management emerge.

Functions of financial management:

Functions of financial management can be broadly divided into two groups.

1. **Executive** functions of financial management, and
2. **Routine** functions of financial management.

This division of functions of financial management is depicted below.

Executive Functions of Financial Management:

The executive functions of financial management are depicted & listed below.

Eight executive functions of financial management (FM) are: -

1. **Estimating capital requirements:** The company must estimate its capital requirements (needs) very carefully. This must be done at the promotion stage. The company must estimate its fixed capital needs and working capital need. If not, the company will become over-capitalized or under-capitalized.
2. **Determining capital structure:** Capital structure is the ratio between owned capital and borrowed capital. There must be a balance between owned capital and borrowed capital. If the company has too much owned capital, then the shareholders will get fewer dividends. Whereas, if the company has too much of borrowed capital, it has to pay a lot of interest. It also has to repay the borrowed capital after some time. So the finance managers must prepare a balanced capital structure.
3. **Estimating cash flow:** Cash flow refers to the cash which comes in and the cash which goes out of the business. The cash comes in mostly from sales. The cash goes out for business expenses. So, the finance manager must estimate the future sales of the business. This is called Sales forecasting. He also has to estimate the future business expenses.
4. **Investment Decisions:** The business gets cash, mainly from sales. It also gets cash from other sources. It gets long-term cash from equity shares, debentures, term loans from financial institutions, etc. It gets short-term loans from banks, fixed deposits, dealer deposits, etc. The finance manager must invest the cash properly. Long-term cash must be used for purchasing fixed assets. Short-term cash must be used as a working capital.
5. **Allocation of surplus:** Surplus means profits earned by the company. When the company has a surplus, it has three options, viz.,

1. It can pay dividend to shareholders.
2. It can save the surplus. That is, it can have retained earnings.
3. It can give bonus to the employees.

6. **Deciding Additional finance:** Sometimes, a company needs additional finance for modernization, expansion, diversification, etc. The finance manager has to decide on following questions.

1. When the additional finance will be needed?
2. For how long will this finance be needed?
3. From which sources to collect this finance?
4. How to repay this finance?

Additional finance can be collected from shares, debentures, loans from financial institutions, fixed deposits from public, etc.

7. **Negotiating for additional finance:** The finance manager has to negotiate for additional finance. That is, he has to speak to many bank managers. He has to persuade and convince them to give loans to his company. There are two types of loans, viz., short term loans and long-term loans. It is easy to get short-term loans from banks. However, it is very difficult to get long-term loans.

8. **Checking the financial performance:** The finance manager has to check the financial performance of the company. This is a very important finance function. It must be done regularly. This will improve the financial performance of the company. Investors will invest their money in the company only if the financial performance is good. The finance manager must compare the financial performance of the company with the established standards. He must find ways for improving the financial performance of the company.

Routine Functions of Financial Management:

The routine functions are also called as **incidental** functions. Routine functions are clerical Functions. They help to perform the Executive functions of financial management. The routine functions of financial management are briefly listed below. Six routine functions of financial management (FM) are: -

1. Supervision of cash receipts and payments.
2. Safeguarding of cash balances.
3. Safeguarding of securities, insurance policies and other valuable papers.
4. Taking proper care of mechanical details of financing.
5. Record keeping and reporting.
6. Credit Management.

- Understanding Capital Markets

Objectives of Financial Management:

- **Profit maximization :** The main objective of financial management is profit maximization. The finance manager tries to earn maximum profits for the company in the short-term and the long-term. He cannot guarantee profits in the long term because of business uncertainties. However, a company can earn maximum profits even in the long term, if:-
- The Finance manager takes proper financial decisions. He uses the finance of the company properly.

- **Wealth maximization:** Wealth maximization (shareholders' value maximization) is also a main objective of financial management. Wealth maximization means to earn maximum wealth for the shareholders. So, the finance manager tries to give a maximum dividend to the shareholders. He also tries to increase the market value of the shares. The market value of the shares is directly related to the performance of the company. Better the performance, higher is the market value of shares and vice-versa. So, the finance manager must try to maximize shareholder's value.
- **Proper estimation of total financial requirements:** Proper estimation of total financial requirements is a very important objective of financial management. The finance manager must estimate the total financial requirements of the company. He must find out how much finance is required to start and run the company. He must find out the fixed capital and working capital requirements of the company. His estimation must be correct. If not, there will be shortage or surplus of finance. Estimating the financial requirements is a very difficult job. The finance manager must consider many factors, such as the type of technology used by company, number of employees employed, scale of operations, legal requirements, etc.
- **Proper mobilization:** Mobilization (collection) of finance is an important objective of financial management. After estimating the financial requirements, the finance manager must decide about the sources of finance. He can collect finance from many sources such as shares, debentures, bank loans, etc. There must be a proper balance between owned finance and borrowed finance. The company must borrow money at a low rate of interest.
- **Proper utilization of finance:** Proper utilization of finance is an important objective of financial management. The finance manager must make optimum utilization of finance. He must use the finance profitably. He must not waste the finance of the company. He must not invest the company's finance in unprofitable projects. He must not block the company's finance in inventories. He must have a short credit period.
- **Maintaining proper cash flow:** Maintaining proper cash flow is a short-term objective of financial management. The company must have a proper cash flow to pay the day-to-day expenses such as purchase of raw materials, payment of wages and salaries, rent, electricity bills, etc. If the company has a good cash flow, it can take advantage of many opportunities such as getting cash discounts on purchases, large-scale purchasing, giving credit to customers, etc. A healthy cash flow improves the chances of survival and success of the company.
- **Survival of company:** Survival is the most important objective of financial management. The company must survive in this competitive business world. The finance manager must be very careful while making financial decisions. One wrong decision can make the company sick, and it will close down.
- **Creating reserves:** One of the objectives of financial management is to create reserves. The company must not distribute the full profit as a dividend to the shareholders. It must keep a part of its profit as reserves. Reserves can be used for future growth and expansion. It can also be used to face contingencies in the future.

- **Proper coordination:** Financial management must try to have proper coordination between the finance department and other departments of the company.
- **Create goodwill:** Financial management must try to create goodwill for the company. It must improve the image and reputation of the company. Goodwill helps the company to survive in the short-term and succeed in the long-term. It also helps the company during bad times.
- **Increase efficiency:** Financial management also tries to increase the efficiency of all the departments of the company. Proper distribution of finance to all the departments will increase the efficiency of the entire company.
- **Financial discipline:** Financial management also tries to create a financial discipline. Financial discipline means:-
 - To invest finance only in productive areas. This will bring high returns (profits) to the company.
 - To avoid wastage and misuse of finance.
- **Reduce cost of capital :** Financial management tries to reduce the cost of capital. That is, it tries to borrow money at a low rate of interest. The finance manager must plan the capital structure in such a way that the cost of capital is minimized.
- **Reduce operating risks :** Financial management also tries to reduce the operating risks. There are many risks and uncertainties in a business. The finance manager must take steps to reduce these risks. He must avoid high-risk projects. He must also take proper insurance.
- **Prepare capital structure :** Financial management also prepares the capital structure. It decides the ratio between owned finance and borrowed finance. It brings a proper balance between the different sources of capital. This balance is necessary for liquidity, economy, flexibility and stability.

Features of Financial Management:

- According to **Solomon**, "Financial management is concerned with the efficient use of an important economic resource, namely, capital funds."
- According to **J. L. Massie**, "Financial management is the operational activity of a business that is responsible for obtaining and effectively utilizing the funds necessary for efficient operation."
- According to **Weston & Brigham**, "Financial management is an area of financial decision making harmonizing individual motives & enterprise goals."
- According to **Howard & Upton**, "Financial management is the application of the planning & control functions of the finance function."
- According to **J. F. Bradley**, "Financial management is the area of business management devoted to the judicious use of capital & careful selection of sources of capital in order to enable a spending unit to move in the direction of reaching its goals."

Main features of financial management:

On the basis of the above definitions, the following are the main characteristics of the financial management-

- **Analytical Thinking**-Under financial management financial problems are analyzed and considered. Study of trend of actual figures is made and ratio analysis is done.

- **Continuous Process**-previously financial management was required rarely but now the financial manager remains busy throughout the year.
- **Basis of Managerial Decisions**- All managerial decisions relating to finance are taken after considering the report prepared by the finance manager. The financial management is the base of managerial decisions.
- **Maintaining Balance between Risk and Profitability**-Larger the risk in the business larger is the expectation of profits. Financial management maintains balance between the risk and profitability.
- **Coordination between Process**- There is always a coordination between various processes of the business.
- **Centralized Nature**- Financial management is of a centralized nature. Other activities can be decentralized but there is only one department for financial management.

Responsibility of Financial Management:

The responsibilities of financial management or financial manager vary widely from one business unit to another, depending upon the size and the nature of the business. In the light of this wide diversity of organizational practices, it is not surprising to find that in most of the company, financial officer is responsible for the routine finance functions. The main responsibilities of the financial officer are as follows:

1. Financial Planning

The main responsibility of the chief financial officer in a large concern is to forecast the needs and sources of finance and ensure the adequate supply of cash at proper time for the smooth running of the business. He is to see that cash inflow and outflow must be uninterrupted and continuous. For this purpose, financial planning is necessary, i.e., he must decide the time when

he needs money, the sources of supply of money and the investment patterns so that the company may meet its obligations properly and maintain its goodwill in the market. The financial manager is also to see that there is no surplus money in the business which earns nothing.

2. Raising of Necessary Funds

The second main responsibility of the financial officer is to see the nature of the need, i.e., whether finances are required for long-term or for short-term. He must assess the alternative sources of supply of finance taking into view the cost of raising funds, its effect on various concerned parties, i.e. shareholders, creditors, employees and the society, control and risk in financing and elasticity in capital structure etc.

3. Controlling the Use of Funds

The financial manager is also responsible for the proper utilization of funds. Assets must be used effectively so as to earn higher profits; inflow and outflow of cash must be controlled in a manner so as to meet the current as well as future obligations; unnecessary expenditure should be curtailed and there should be left no possibility for misappropriation of money.

4. Disposition of profits

Appropriation of profits is one of the main responsibilities of the financial manager. He is to advise to the top executives as how much of the profits should be retained in the business as reserves for future expansion; how much to be used in repaying the debts; and how much to be distributed to the shareholders as dividend. On the basis of the advice given by the financial manager, the resolutions regarding depreciations, reserves, general reserves and distribution of dividends are carried out in the meeting of board of directors of the company.

5. Other Responsibilities

Over and above, the responsibilities stated above, there are certain other responsibilities of the financial manager. These are:

a) Responsibility to owners

Shareholders or stock-holders are the real owners of the concern. Financial manager has the prime responsibility to those who have committed funds to the enterprise. He should not only maintain the financial health of the enterprise, but should also help to produce a rate of earning that will reward the owners adequately for the risk capital they provide.

b) Legal obligations

Financial manager is also under an obligation to consider the enterprise in the light of its legal obligations. A host of laws, taxes and rules and regulations cover nearly every move and policy. Good financial management helps to develop a sound legal framework.

c) Responsibilities of Employees

The financial management must try to produce a healthy going concern capable of maintaining regular employment at satisfactory rate of pay under favourable working conditions. The long term financial interests of management, employee's, owners are common.

d) Responsibilities to customers

In order to make the payments of its customers' bill, the effective financial management is necessary. Sound financial management ensures the creditors continued supply of raw material.

e) Wealth Maximization

Prof. Solomon of Stanford University has argued that the main goal of the finance function is wealth maximization. The other goals may be achieved automatically. In the light of the above discussion, we can conclude that the main responsibility of the financial manager is not only to maintain the financial health of the organization but also to increase the economic welfare of the shareholders by utilizing the funds in an effective manner.

Scope of financial management in any organization of your choice:

Scope of Financial Management:

Financial management has a wide scope. According to Dr. S. C. Saxena, the scope of financial management includes the following five 'A's.

Anticipation : Financial management estimates the financial needs of the company. That is, it finds out how much finance is required by the company.

Acquisition : It collects finance for the company from different sources.

Allocation : It uses this collected finance to purchase fixed and current assets for the company.

Appropriation : It divides the company's profits among the shareholders, debenture holders, etc. It keeps a part of the profits as reserves.

Assessment : It also controls all the financial activities of the company. Financial management is the most important functional area of management. All other functional areas such as production management, marketing management, personnel management, etc. depends on Financial management. Efficient financial management is required for survival, growth and success of the company or firm.

“The goal of profit maximization does not provide an operationally useful criterion” –

Explain:

Profit maximization refers to the maximization of income or earnings of a firm. The arguments in favour of profit maximization as the objectives of financial management are

1. **Nature Goal :** Profit is the aim of any business. Naturally, the goal of financial management should be profit maximization.

2. **Measure of Efficiency :** Profit is a measure of efficiency. Higher profits imply greater efficiently. Hence, the objective of profit maximization is quite rational.

3. **Internal Generation of Funds** Profit lead to internal generation of funds. It helps to finance the growth of the business.

4. **Protection against Risks :** Profits provide protection against risks when a company is faced with unfavourable conditions, (such as fall in prices, increase n costs and severe competition). Accumulated profits serve as a cushion to absorb the shocks.

5. **Fulfillment of Social Obligations:** Profits are essential for fulfilling social obligations of the business. The goal or profit maximization helps to maximize social welfare.

1. **Vague :** The term profit is vague. It has different meanings. For instance, profit may refer to long-term profits or short-term profits. It may refer to profit before tax or profit after tax or short-term profits. It may refer to profit before tax or profit after tax or even operating profits.

2. **Neglects Time Value of Money :** The objective of profit maximization neglects time value of money. Profits of today are more valuable than profits to be earned after five years. But profit maximization objective treats all profits as equal, irrespective of the timing.

3. **Ignores Risk Factor.** Some projects are more risky than the others though the expected earnings may be equal. But, the risk factor is not considered by the profit maximization goal.

4. **Taint of Immorality:** Profit maximization implies exploitation of consumers, workers and the society. Hence, it is regarded as immoral.

5. **Invalid:** Profit maximization may be a valid objective under conditions of perfect competition. As the markets are not perfect, it cannot be a valid objective.

6. **Inadequate:** In company form of a organization, there is separation of ownership and control. Shareholders are the owners. But, control is in the hands of professional managers.

Creditors, financial institutions, workers, consumers and the society are concerned with the company's operations. The management has to reconcile the conflicting interests of these stake holders. Profit maximization goal is inadequate for the purpose.

IMPORTANCE OF FINANCIAL MANAGEMENT:

Finance is the lifeblood of business organization. It needs to meet the requirement of the business concern. Each and every business concern must maintain adequate amount of finance for their smooth running of the business concern and also maintain the business carefully to achieve the goal of the business concern. The business goal can be achieved only with the help of effective management of finance. We can't neglect the importance of finance at any time at and at any situation. Some of the importance of the financial management is as follows:

Financial Planning

Financial management helps to determine the financial requirement of the business concern and leads to take financial planning of the concern. Financial planning is an important part of the business concern, which helps to promotion of an enterprise.

Acquisition of Funds

Financial management involves the acquisition of required finance to the business concern. Acquiring needed funds play a major part of the financial management, which involve possible source of finance at minimum cost.

Proper Use of Funds

Proper use and allocation of funds leads to improve the operational efficiency of the business concern. When the finance manager uses the funds properly, they can reduce the cost of capital and increase the value of the firm.

Financial Decision

Financial management helps to take sound financial decision in the business concern. Financial decision will affect the entire business operation of the concern. Because there is a direct relationship with various department functions such as marketing, production personnel, etc.

Improve Profitability

Profitability of the concern purely depends on the effectiveness and proper utilization of funds by the business concern. Financial management helps to improve the profitability position of the concern with the help of strong financial control devices such as budgetary control, ratio analysis and cost volume profit analysis.

Increase the Value of the Firm

Financial management is very important in the field of increasing the wealth of the investors and the business concern. Ultimate aim of any business concern will achieve the maximum profit and higher profitability leads to maximize the wealth of the investors as well as the nation.

Promoting Savings

Savings are possible only when the business concern earns higher profitability and maximizing wealth. Effective financial management helps to promoting and mobilizing individual and corporate savings. Nowadays financial management is also popularly known as business finance or corporate finances. The business concern or corporate sectors cannot function without the importance of the financial management.

The variants of profit maximization goal are :

To overcome the problem of vagueness, the term profit may be defined specifically as profit after taxes. But this objective is also not sound as it will not maximize the economic welfare of the shareholders.

For example, A Ltd has 10,000 equity and earns a profit of Rs.50,000. The earnings per share (EPS) are Rs.5. The company issues 5000 additional shares and invests the proceeds in the business. If profit after tax increases to Rs.60,000 the EPS will decline to Rs.4 (Rs.60,000 divided by 15,000). It is clear that maximizing profit after tax is not necessarily advantageous to the share holders.

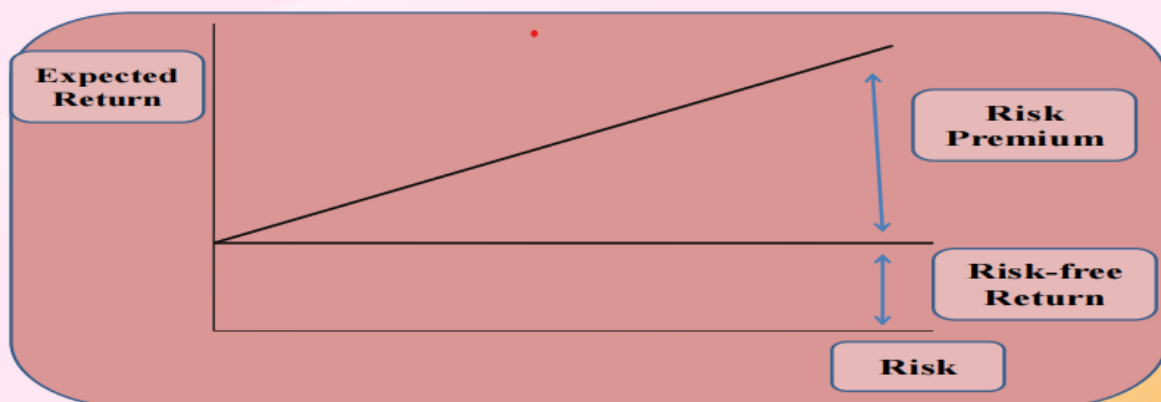
ii) Maximizing the EPS

The objective of maximization of earnings per share also suffers from draw backs. It ignores time value of money and the risk element. Because of the drawbacks profit maximization, as an objective of financial management has been rejected.

Risk-return Trade-off

- Risk and expected return move in tandem; the greater the risk, the greater the expected return.
- Financial decisions of the firm are guided by the **risk-return trade-off**.
- The return and risk relationship: $\text{Return} = \text{Risk-free rate} + \text{Risk premium}$
- Risk-free rate is a compensation for time and risk premium for risk.

• The Risk-return Relationship



• An overview of (functions)Financial Management



Managers versus Shareholders' Goals

- A company has stakeholders such as employees, debt-holders, consumers, suppliers, government and society.
- Managers may perceive their role as reconciling conflicting objectives of stakeholders. This stakeholders' view of managers' role may compromise with the objective of SWM.
- Managers may pursue their own personal goals at the cost of shareholders, or may play safe and create satisfactory wealth for shareholders than the maximum.
- Managers may avoid taking high investment and financing risks that may otherwise be needed to maximize shareholders' wealth. Such "satisfying" behaviour of managers will frustrate the objective of SWM as a normative guide.

Financial Goals and Firm's Mission and Objectives

- Firms' primary objective is maximizing the welfare of owners, but, in operational terms, they focus on the satisfaction of its customers through the production of goods and services needed by them
- Firms state their vision, mission and values in broad terms
- Wealth maximization is more appropriately a decision criterion, rather than an objective or a goal.
- Goals or objectives are missions or basic purposes of a firm's existence
- The shareholders' wealth maximization is the second-level criterion ensuring that the decision meets the minimum standard of the economic performance.
- In the final decision-making, the judgement of management plays the crucial role. The wealth maximization criterion would simply indicate whether an action is economically viable or not.

Organisation of the Finance Functions

- Reason for placing the finance functions in the hands of top management
 - Financial decisions are crucial for the survival of the firm.
 - The financial actions determine solvency of the firm
 - Centralisation of the finance functions can result in a number of economies to the firm.

Status and Duties of Finance Executives

- The exact organisation structure for financial management will differ across firms.
- The financial officer may be known as the financial manager in some organisations, while in others as the vice-president of finance or the director of finance or the financial controller.

Role of Treasurer and Controller

- Two more officers—the **treasurer** and the **controller**—may be appointed under the direct supervision of CFO to assist him or her.
- The treasurer's function is to raise and manage company funds while the controller oversees whether funds are correctly applied.

Time Preference for Money

- Most individuals value the opportunity to receive money now higher than waiting for one or more periods to receive the same amount.

- **Time preference for money** is an individual's preference for possession of a given amount of money now, rather than the same amount at some future time.

- Three reasons may be attributed to the individual's time preference for money:

- risk
- preference for consumption
- investment opportunities

Required Rate of Return

- The time preference for money is generally expressed by an interest rate. This rate will be positive even in the absence of any risk. It may be therefore called the **risk-free rate**.

- An investor requires compensation for assuming risk, which is called **risk premium**.

- The investor's **required rate of return** is:

Risk-free rate + Risk premium

➤ **The Required rate of return may also be called the opportunity cost of capital Time Value Adjustment**

- Two most common methods of adjusting cash flows for time value of money:

- **Compounding**—the process of calculating **future values** of cash flows and
- **Discounting**—the process of calculating **present values** of cash flows.

Future Value

- **Compounding** is the process of finding the future values of cash flows by applying the concept of compound interest.

- **Compound interest** is the interest that is received on the original amount (principal) as well as on any interest earned but not withdrawn during earlier periods.

- **Simple interest** is the interest that is calculated only on the original amount (principal), and thus, no compounding of interest takes place.

Present Value

- **Present value** of a future cash flow (inflow or outflow) is the amount of current cash that is of equivalent value to the decision-maker.

- **Discounting** is the process of determining present value of a series of future cash flows.

- The interest rate used for discounting cash flows is also called the discount rate.

Generally individuals show a time preference for money. Give reasons for such a preference:

The concept of time value of money is simple. According to this concept, the same amount of cash, receivable during different time periods has different values. The value of money received today is greater than the value of the same amount receivable after 5 to 10 years. The sooner the money is received the better it is. If Mr. Rajesh is given the option of receiving Rs.10,000 today or after one year he will definitely prefer to receive the amount, today itself. This is called time preference for money. However, he may be ready to receive the money after one year. If he is suitably rewarded for his waiting. The reward is called interest. Mrs. Rajesh may agree to receive Rs.10,000 after 1 year if he is paid an interest of Rs.1200. So, time preference for money is expressed as interest rate. Hence, time value of money is to be recognized in making financial decisions. This is done by making appropriate adjustments through

discounting or compounding of cash flows. Reasons for Time Preference for Money People prefer to receive money, earlier than later. The reasons are :

1. **Uncertainty:** Future is uncertain. There is a chance of not getting the money at all. Hence, people like to receive the money today itself rather than waiting for the future.
2. **Preference for Consumption:** The money may be needed to meet urgent current needs. Therefore, people prefer to receive money as early as possible.
3. **Investment Opportunities:** Money has time value. If Mr. Chandran received Rs.50,000 today, he can invest the amount and earn interest. Suppose he gets an interest of 10% he will have Rs.55,000 at the end of 1 year. Therefore, it is good for him to receive Rs.50,000 now as it will grow into Rs.55,000 after one year.

What is annuity? Explain with an example explain how can future value of an annuity be determined:

"A Rupee today is better than a Rupee tomorrow" The earlier the money is received, greater the potential for increasing the wealth. Due to inflation, money may lose its purchasing power overtime.

$$\text{Present Value Factor (PVF)} = 1 / (1 + r)^n$$

$$\text{Present value (PV)} = \text{Cash flows} * \text{PVF}$$

Illustration

What is the present value of Rs. 10,000 to be received after 5 years when the rate of interest is 10%?

Solution:

$$\text{PVF} = 1 / (1 + r)^n = 1 / (1 + 0.10)^5 = 0.621$$

$$\text{PV} = 10,000 * 0.621 = 6,210$$

Illustration

A Rs. 10,000 par value bond bearing a coupon rate of 12% will mature after 5 years. **What is the value of the bond**, if the discount rate is 15%?

Solution:

$$\text{Interest earned p.a.} = 10,000 * 12/100 = \text{Rs. } 1,200$$

$$\text{Principal} = \text{Rs. } 10,000$$

$$\text{PVA} = (1 - [1 / (1 + r)^n]) / r$$

$$\text{PVA} = \{1 - [1 / (1 + 0.15)^5] / 0.15\} = 3.352$$

$$\text{PVF} = 1 / (1 + r)^n = 1 / (1 + 0.15)^5 = 0.497$$

$$\begin{aligned} \text{Value of bond} &= (3.352 * 1,200) + (0.497 * 10,000) \\ &= 4,022 + 4970 = \text{Rs. } 8,992 \end{aligned}$$

Relationship between effective rate of interest and nominal rate of interest:

The nominal interest rate is the periodic interest rate times the number of periods per year. For example, a nominal annual interest rate of 12% based on monthly compounding means a 1% interest rate per month (compounded). A nominal interest rate for compounding periods less than a year is always lower than the equivalent rate with annual compounding (this immediately follows from elementary algebraic manipulations of the formula for compound

interest). Note that a nominal rate without the compounding frequency is not fully defined: for any interest rate, the effective interest rate cannot be specified without knowing the compounding frequency and the rate. Although some conventions are used where the compounding frequency is understood, consumers in particular may fail to understand the importance of knowing the effective rate.

Nominal interest rates are not comparable unless their compounding periods are the same; effective interest rates correct for this by "converting" nominal rates into annual compound interest. In many cases, depending on local regulations, interest rates as quoted by lenders and in advertisements are based on nominal, not effective interest rates, and hence may understate the interest rate compared to the equivalent effective annual rate. The term should not be confused with simple interest (as opposed to compound interest) which is not compounded.

The effective interest rate is always calculated as if compounded annually. The effective rate is calculated in the following way, where r is the effective rate, i the nominal rate (as a decimal, e.g. 12% = 0.12), and n the number of compounding periods per year (for example, 12 for monthly compounding):

Monthly compounding:

Example 1: A nominal interest rate of 6%/a compounded monthly is equivalent to an effective interest rate of 6.17%.

Example 2: 6% annually is credited as $6\%/12 = 0.5\%$ every month. After one year, the initial capital is increased by the factor $(1+0.005)^{12} \approx 1.0617$.

Daily compounding:

A loan with daily compounding will have a substantially higher rate in effective annual terms. For a loan with a 10% nominal annual rate and daily compounding, the effective annual rate is **Solution:**

$$PVF = 1 / (1 + r)^n = 1 / (1 + 0.10)^5 = 0.621$$

$$PV = 10,000 * 0.621 = 6,210$$

Illustration

A Rs. 10,000 par value bond bearing a coupon rate of 12% will mature after 5 years. **What is the value of the bond**, if the discount rate is 15%?

Solution:

$$\text{Interest earned p.a.} = 10,000 * 12/100 = \text{Rs. } 1,200$$

$$\text{Principal} = \text{Rs. } 10,000$$

$$PVAF = (1 - [1 / (1 + r)^n]) / r$$

$$PVAF = \{1 - [1 / (1 + 0.15)^5] / 0.15\} = 3.352$$

$$PVF = 1 / (1 + r)^n = 1 / (1 + 0.15)^5 = 0.497$$

$$\text{Value of bond} = (3.352 * 1,200) + (0.497 * 10,000) = 4,022 + 4,970 = \text{Rs. } 8,992$$

Relationship between effective rate of interest and nominal rate of interest:

The nominal interest rate is the periodic interest rate times the number of periods per

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$$r = (1 + i/n)^n - 1$$

Monthly compounding:

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Example 2: 6% annually is credited as 6%/12 = 0.5% every month. After one year, the initial capital is increased by the factor $(1+0.005)^{12} \approx 1.0617$.

Daily compounding:

A loan with daily compounding will have a substantially higher rate in effective annual terms. For a loan with a 10% nominal annual rate and daily compounding, the effective annual rate is 10.516%. For a loan of \$10,000 (paid at the end of the year in a single lump sum), the borrower would pay \$51.56 more than one who was charged 10% interest, compounded annually.

“Financial management is nothing but managerial decision making on asset mix, capital mix and profit allocation” – Explain:

a. Investment Decision

One of the most important finance functions is to intelligently allocate capital to long term assets. This activity is also known as capital budgeting. It is important to allocate capital in those long term assets so as to get maximum yield in future. Following are the two aspects of investment decision

- Evaluation of new investment in terms of profitability
- Comparison of cut off rate against new investment and prevailing investment.

Since the future is uncertain therefore there are difficulties in calculation of expected return. Along with uncertainty comes the risk factor which has to be taken into consideration. This risk factor plays a very significant role in calculating the expected return of the prospective investment. Therefore while considering investment proposal it is important to take into consideration both expected return and the risk involved.

b. Financial Decision

Financial decision is yet another important function which a financial manager must perform. It is important to make wise decisions about when, where and how should a business acquire funds. Funds can be acquired through many ways and channels. Broadly speaking a correct ratio of an equity and debt has to be maintained. This mix of equity capital and debt is known as a firm's capital structure. A firm tends to benefit most when the market value of a company's share maximizes this not only is a sign of growth for the firm but also maximizes shareholders wealth. On the other hand the use of debt affects the risk and return of a shareholder. It is more risky though it may increase the return on equity funds. A sound financial structure is said to be one which aims at maximizing shareholders return with minimum risk. In such a scenario the market value of the firm will maximize and hence an optimum capital structure would be achieved. Other than equity and debt there are several other tools which are used in deciding a firm capital structure.

c. Dividend Decision

Earning profit or a positive return is a common aim of all the businesses. But the key function a financial manager performs in case of profitability is to decide whether to distribute all the profits to the shareholder or retain all the profits or distribute part of the profits to the shareholder and retain the other half in the business. It's the financial manager's responsibility to decide a optimum dividend policy which maximizes the market value of the firm. Hence an optimum dividend payout ratio is calculated. It is a common practice to pay regular dividends in case of profitability another way is to issue bonus shares to existing shareholders.

d. Liquidity Decision

It is very important to maintain a liquidity position of a firm to avoid insolvency. Firm's profitability, liquidity and risk all are associated with the investment in current assets. In order to maintain a tradeoff between profitability and liquidity it is important to invest sufficient funds in current assets. But since current assets do not earn anything for business therefore a proper calculation must be done before investing in current assets. Current assets should properly be valued and disposed of from time to time once they become non profitable. Current assets must be used in times of liquidity problems and times of insolvency

Options

- An option is a contract that gives the holder a right, without any obligation, to buy or sell an asset at an agreed price on or before a specified period of time.
- The option to buy an asset is known as a **call option**.
- The option to sell an asset is called a **put option**.
- The price at which option can be exercised is called an exercise **price** or a strike price.

- The asset on which the put or call option is created is referred to as the **underlying asset**.
- The **option premium** is price that the holder of an option has to pay for obtaining a call or a put option.

When an Option can be exercised

- **European option** When an option is allowed to be exercised only on the maturity date, it is called a European option.
- **American option** When the option can be exercised any time before its maturity, it is called an American option.

Possibilities at Expiration

- **In-the-money** A put or a call option is said to in-the-money when it is advantageous for the investor to exercise it.
- **Out-of-the-money** A put or a call option is out-of-the-money if it is not advantageous for the investor to exercise it.
- **At-the-money** When the holder of a put or a call option does not lose or gain whether or not he exercises his option.

Call Option

Buy a call option

- a. You should exercise call option when:

Share price at expiration > Exercise price.

- b. Do not exercise call option when:

Share price at expiration < Exercise price.

- c. The value of the call option at expiration is:

Value of call option at expiration = Maximum

[Share price – Exercise price, 0].

- d. The expression above indicates that the value of a call option at expiration is the maximum of the share price minus the exercise price or zero.

The call buyer's gain is call seller's loss.

Put Option

Buy a put option

- o Exercise the put option when:

▪ Exercise price > Share price at expiration.

- o Do not exercise the put option when:

▪ Exercise price < Share price at expiration.

- o The value or payoff of a put option at expiration will be:

Value of put option at expiration = Maximum [Exercise price – Share price at expiration, 0].

The put option buyer's gain is the seller's loss.

Options Trading in India

- i. The Security Exchange Board of India (SEBI) has announced a list of 31 shares for the stock-based option trading from July 2002. SEBI selected these shares for option trading on the basis of the following criteria:
- ii. Shares must be among the top 200 in terms of market capitalisation and trading volume.

- iii. Shares must be traded in at least 90 per cent of the trading days.
- iv. The non-promoter holding should be at least 30 per cent and the market capitalisation of free-float shares should be Rs 750 crore.
- v. The six-month average trading volume in the share in the underlying cash market should be a minimum of Rs 5 crore.
- vi. The ratio of daily volatility of the share vis-à-vis the daily volatility of the index should not be more than four times at any time during the previous six months.
- vii. The minimum size of the contract is Rs 2 lakh. For the first six months, there would be cash settlement in options contracts and afterwards, there would be physical settlement. The option sellers will have to pay the margin, but the buyers will have to only pay the premium in advance. The stock exchanges can set limits on exercise price.

Index Options

- i. Index options are call or put options on the stock market indices. In India, there are options on the Bombay Stock Exchange (BSE)—Sensex and the National Stock Exchange (NSE)—Nifty.
- ii. The Sensex options are European-type options and expire on the last Thursday of the contract month. The put and call index option contracts with 1-month, 2-month and 3-month maturity are available. The settlement is done in cash on a T + 1 basis and the prices are based on expiration price as may be decided by the Exchange. Option contracts will have a multiplier of 100.
- iii. The multiplier for the NSE Nifty Options is 200 with a minimum price change of Rs 10 (200×0.05).

Factors Determining Option Value

- 1. Exercise price and the share (underlying asset) price
- 2. Volatility of returns on share
- 3. Time to expiration
- 4. Interest rates

Model for Option Valuation

- Simple binomial tree approach to option valuation.
- Black-Scholes option valuation model.

Simple Binomial Tree Approach

- Sell a call option on the share. We can create a portfolio of certain number of shares (let us call it delta, D) and one call option by going long on shares and short on options that there is no uncertainty of the value of portfolio at the end of one year.
- Formula for determining the **option delta**, represented by symbol D, can be written as follows:
 - Option Delta = Difference in option Values / Difference in Share Prices.
 - The value of portfolio at the end of one year remains same irrespective of the increase or decrease in the share price.
 - The value of the call option will remain the same irrespective of any probabilities of increase or decrease in the share price. This is so because the option is valued in terms of the price of the underlying share, and the share price already includes the probabilities of its rise or fall.

Black and Scholes Model for Option Valuation

- ❖ The B–S model is based on the following assumptions:
- ❖ The rates of return on a share are log normally distributed.
- ❖ The value of the share (the underlying asset) and the risk-free rate are constant during the life of the option.
- ❖ The market is efficient and there are no transaction costs and taxes.
- ❖ There is no dividend to be paid on the share during the life of the option.
- ❖ The B–S model is as follows: $C_0 = S_0 N(d_1) - Ee^{-rft} N(d_2)$
- ❖ where
- ❖ C_0 = the current value of call option
- ❖ S_0 = the current market value of the share
- ❖ E = the exercise price
- ❖ $e = 2.7183$, the exponential constant
- ❖ r_f = the risk-free rate of interest
- ❖ t = the time to expiration (in years)
- ❖ $N(d_1)$ = the cumulative normal probability density function

Features of B–S Model

- Black–Scholes model has two features-

- ❖ The parameters of the model, except the share price volatility, are contained in the agreement between the option buyer and seller.
- ❖ In spite of its unrealistic assumptions, the model is able to predict the true price of option reasonably well.

The model is applicable to both European and American options with a few adjustments

Option's Delta or Hedge Ratio

- The **hedge ratio** is a tool that enables us to summaries the overall exposure of portfolios of options with various exercise prices and maturity periods.
- An option's hedge ratio is the change in the option price for a Re 1 increase in the share price.
- A call option has a positive hedge ratio and a put option has a negative hedge ratio.
- Under the Black–Scholes option valuation formula, the hedge ratio of a call option is $N(d_1)$ and the hedge ratio for a put is $N(d_1) - 1$.

Dividend-Paying Share Option

- We can use slightly modified

B–S model for this purpose. The share price will go down by an amount reflecting the payment of dividend. As a consequence, the value of a call option will decrease and the value of a put option will increase.

- We also need to adjust the volatility in case of a dividend-paying share since in the B–S model it is the volatility of the risky part of the share price. This is generally ignored in practice.

Ordinary Share as an Option

- The limited liability feature provides an opportunity to the shareholders to default on a debt.

■ The debt-holders are the sellers of call option to the shareholders. The amount of debt to be repaid is the exercise price and the maturity of debt is the time to expiration.

■ The shareholders' option can be interpreted as a put option. The shareholders can sell (hand-over) the firm to the debt-holders at zero exercise price if they do not want to make the payment that is due.

PROBLEMS & SOLUTIONS

1. A patent has been purchased for Rs.17,50,000 has a remaining life of 13 years and Rs.2,50,000 salvage value. It is estimated that the patent will generate operating revenues Rs.3,50,000 per year throughout its life. Operating cost will be Rs.80,000 per year from year 2 to 13. Using an interest rate of 12% p.a. find the present values of this investment. **(May/June 2007)**

Solution:

CIF = Revenue – cost

$$3,50,000 - 80,000 = 2,70,000 \text{ (PVF for 13 years)}$$

$$= 2,70,000$$

Present Value of CIF = 2,70,000 * PVF (2-13 years)

Present Value of Salvage value = 2,50,000 * PVF @12%

Net present value = Present value of CIF + Present Value of Salvage Value

$$= 17,50,000$$

2. The required rate of return on a portfolio with beta of 1.2 is 18% and the risk-free is 6%.

According to CAPM:

1. What is the expected rate of return on the market portfolio?

2. A stock, say delta, with beta of 1.5, sells for Rs.50, one year from now it is expected to yield a dividend income of Rs.6. What price do investors expect after one year?

(May/June 2007)

Solution:

$$1) \text{ Expected return} = R_f + \beta(R_m - R_f)$$

$$18\% = 6\% + 1.2(R_m - 6\%)$$

$$R_m = 16\%$$

$$2) \text{ Expected return after one year:}$$

$$= 6\% + (16\% - 6\%)$$

$$= 6\% + 10\% = 16\%$$

3. On 31st December 1980, the balance sheet of a limited company disclosed the following position: **(Nov/Dec 2007)**

Balance Sheet as on 31.12.1980

Equity Share Capital (40,000 equity shares of Rs.10 each fully paid)	4,00,000	Goodwill	40,000
		Fixed Assets	5,00,000
General reserve	90,000	Current assets	2,00,000
P/L account	20,000		
5% Debentures	1,00,000		
Current Liabilities	1,30,000		
	7,40,000		7,40,000

On 31st December 1970, the fixed assets were independently valued at Rs.5,50,000 and the good will at Rs. 50,000. The net profits for the last three years were as under

1968	Rs.51,600
1969	Rs.52,000
1970	Rs.51,650

Of which 20% was placed to reserve, this proportion being considered reasonable in the industry in which this company is engaged and where a fair return is estimated at the rate of 10%. Calculate the value of the company's share by a

- The assets method and
- The yield method

Solution:

Assets Method:

Goodwill as revalued	Rs.50,000
Tangible assets	Rs. 5,50,000
Current Assets as per Balance sheet	Rs. 2,00,000
	8,00,000
Less: 5% Debentures	Rs. 1,00,000
Current Liabilities	Rs. 1,30,000
	2,30,000
	5,70,000

$$\text{Value per share} = 5,70,000 / 40,000 = \text{Rs. } 14.25$$

Yield Method:

Total Profit	=	51,600+52,000+51,650	= 1,55,250
Average profit	=	1,55,250/3	= 51,750
Less: Transfer to General reserve @ 20%			10,350
			41,400

Expected rate of return on Equity capital

$$= 41,400 / 4,00,000 * 100$$

$$= 10.35\%$$

$$\text{Value per share} = \frac{\text{Expected Rate of return}}{\text{Normal rate}} * \text{paid up value per share}$$

$$\text{Value per share} = 10.35 / 10 * 10 = \text{Rs. } 10.35$$

4. A share is quoted is Rs.60. An investor expects the company to pay a dividend of Rs.3 per share, one year from now. The expected price one year from now is Rs.78.50

Share price at present	=	Rs.60
Expected Dividend	=	Rs.3 per share
Expected share price one year from now	=	Rs.78.50

i. What is the expected dividend yield, the rate of price change and holding period yield?

$$PO = D1 / 1+ke + P1 / 1+ke$$

$$60 = 3 / 1+ke + 78.50 / 1+ke$$

$$60 = 81.50 / 1+ke$$

$$60 + 60 ke = 81.50$$

$$K_e = 35.83\%$$

ii. If the beta of the share is 1.5, the risk free rate is 6 percent and the market risk premium is 10 percent, what is the required rate of return?

$$\text{Risk Free Rate} = R_f = 6\%$$

$$\text{Market Risk Premium} = R_m = 10\%$$

$$\text{Beta of the share} = \beta = 1.5$$

$$\text{Required Rate of return} = R_i = R_f + (R_m - R_f)\beta$$

$$= 6 + (10 - 6) \times 1.5$$

$$= 6 + 6 = 12\%$$

5. Using the following details, calculate, i) expected rate of returns of portfolio in each using capital asset pricing model, ii) average return of portfolio. (May/June 2009)

Investment in	Initial Price	Dividends	Market Price at the end of the	Beta factor
	Rs.	Rs.	Rs.	year
A) Equity share of L&T Ltd.	250	20	500	0.8
H&T Ltd	350	20	600	0.7
P&G Ltd.	450	20	1350	0.5
B) 14% GOI Bonds risk free rate of return 14%	1000		1005	0.99

Solution: i) Calculation of Expected Rate or returns of Portfolio using Capital Asset Pricing Mode :

$$= \frac{\text{Dividend Earned} + \text{Capital Appreciation}}{\text{Initial Investment}} \times 100$$

$$= \frac{(20+20+20+140) + (3455 - 2050)}{(250+350+450+1000)} \times 100$$

$$= \frac{(200 + 1405)}{2050} \times 100$$

$$= 78.29\%$$

By using CAPM Formula :

$$K_e = R_f + \beta (R_m - R_f)$$

$$R_f = \text{Risk Free Rate of Return} = 14\%$$

$$R_m = \text{Market Return} = 78.29\%$$

$$\beta = \text{Beta Factor}$$

$$\text{L\&T} = 14\% + 0.8 (78.29\% - 14\%) = 65.43\%$$

$$\text{H\&L} = 14\% + 0.7 (78.29\% - 14\%) = 59.00\%$$

$$\text{P\&G} = 14\% + 0.5 (78.29\% - 14\%) = 46.145\%$$

$$14\% \text{ GOI} = 14\% + 0.99 (78.29\% - 14\%) = 77.65\%$$

$$\text{ii). Average Return of Portfolio} = \frac{(65.43 + 59 + 46.145 + 77.65)}{4} = 62.06\%.$$

Options:

- An option is a contract that gives the holder a right, without any obligation, to buy or sell an asset at an agreed price on or before a specified period of time.
- The option to buy an asset is known as a **call option**.
- The option to sell an asset is called a **put option**.
- The price at which option can be exercised is called an exercise **price** or a strike price.
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g. The value of the call option at expiration is:

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Options Trading in India

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x. Shares must be traded in at least 90 per cent of the trading days.

xi. The non-promoter holding should be at least 30 per cent and the market capitalisation of free-float shares should be Rs 750 crore.

xii. The six-month average trading volume in the share in the underlying cash market should be a minimum of Rs 5 crore.

xiii. The ratio of daily volatility of the share vis-à-vis the daily volatility of the index should not be more than four times at any time during the previous six months.

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- ❖ There is no dividend to be paid on the share during the life of the option.

o The B–S model is as follows: $C = S_0 N(d_1) - Ee^{-rft} N(d_2)$

- ❖ Where C_0 = the current value of call option
- ❖ S_0 = the current market value of the share
- ❖ E = the exercise price
- ❖ $e = 2.7183$, the exponential constant
- ❖ r_f = the risk-free rate of interest
- ❖ t = the time to expiration (in years)
- ❖ $N(d_1)$ = the cumulative normal probability density function

UNIT-II**INVESTMENT DECISIONS:**

Capital Budgeting: Principles and techniques - Nature of capital budgeting- Identifying relevant cash flows - Evaluation Techniques: Payback, Accounting rate of return, Net Present Value, Internal Rate of Return, Profitability Index - Comparison of DCF techniques - Project selection under capital rationing - Inflation and capital budgeting - Concept and measurement of cost of capital - Specific cost and overall cost of capital

Capital budgeting proposal:

Traditional Techniques: These techniques are generally very simple and easily understandable. But the main drawback of these techniques is that they don't consider the time value of money. But in many industries where an instant decision is to be taken, these methods offer the quicker way out. There are mainly two techniques under this category of methods. They are – Accounting rate of return and Payback period.

Accounting rate of return (ARR):

This method relies on the rate of return each project will earn over its life. It takes the help of accounting profit while calculating the returns. There are 2 methods of calculating ARR

(i) On the basis of original investment,

$$ARR = \frac{\text{average after tax annual net profit}}{\text{Original investment}}$$

This method of calculation was rejected on the ground that the original outlay is gradually recovered over the project life because of depreciation charge.

(ii) On the basis of average investment,

$$ARR = \frac{\text{average annual net profit}}{\text{Original investment} / 2}$$

When depreciation is to be taken on a straight-line basis and no salvage value is assumed, the average investment is always equal to one-half of the original investment, and the resulting rate of return is always twice the rate determined on the basis of original investment.

Advantages of ARR:

- It is easy to understand and simple to calculate
- With the help of this method, direct comparisons among proposed projects of varying lives without a built-in-prejudice in favor of short-term ventures can be made.

Disadvantages of ARR:

- This method ignores time value of money.
- It fails to shed light on yearly rate of return of the project. It may be possible for the project producing higher earnings in the early years to show a lower average rate of return and be rejected in favor of other projects.
- Serious errors can occur in selection of projects if corporate managers accept projects whose accounting rates are equal to or above some arbitrarily selected

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https://play.google.com/store/apps/details?id=info.therithal.brainkart.annauniversitynotes&hl=en_IN**www.BrainKart.com** cut-off rate, and they reject projects whose accounting rates fall short of the cut off rate.

- Accounting information is not suitable for investment decision because it fails to distinguish between cash flowing in and out of the company and book keeping transactions.
- There is no full agreement on the proper measure of the term investment. Thus, different managers have different meanings when they refer to ARR.

Payback period (PBP):

This is the most popular method employed by industrial practitioners for ranking investment projects. This is defined as the “period required for a proposal’s initial cash outlay to be recovered by future additional cash savings generated from the proposal”. The cash flow (after tax & depreciation) is used in calculating the payback period.

$$PBP = CO/CF$$

Where CO = cash outflow of the project and CF = cash inflow

When the cash gains generated by the project are unevenly distributed, cumulative cash gains resulting from the project are to be calculated until the year in which the running total is equal to the amount of investment outlay.

Accept-Reject Rule:

The decision rule is to accept the project if the computed payback period is less than the standard. Otherwise, reject it. While ranking the projects, project with shortest payback period is assigned the highest rank.

Advantages:

- It is easy to understand and calculate
- With the help of this method, projects can be ranked in terms of their economic merits without much of complication.
- It indirectly considers factors like obsolescence and liquidity of investments because project with shortest payback period is exposed to fewer risks.
- This method is useful to the company experiencing shortage of cash because it helps in choosing a project that will yield a quick return of cash fund regardless of its long-term profitability.

Disadvantages:

- It does not measure the profitability of the projects.
- It fails to consider any receipts after the pay-back period, no matter how great they might be. As a result, a project with shorter payback period may be selected against a project with a longer pay back period but longer income producing life and greater return on investment.
- It ignores time value of money. A survey conducted by the Machinery and Allied Products Institute disclosed that about 2/3 rd of the American companies employ pay back approach to appraise merit of projects. This is because, in countries where technological changes are rapid, companies are usually exposed to greater obsolescence risks and where uncertainty surrounding the outcome estimates is great, prime consideration is the speed of capital investment recovery.

Modern / Discounting Cash Flow Techniques: These techniques usually are of more use to businesses in their investment decisions. They take into account the time value of money and adjust their cash flows accordingly before taking a decision. That is the reason why they are

considered superior to the traditional techniques. There are four techniques under this category of methods. They are –

- Net present value (NPV)
- Internal rate of return (IRR)
- Profitability index (PI)
- Discounted Payback Period (DPBP)

Net Present Value (NPV):

In this method, future cash flows are discounted to the present and then compared with the investment outlay. The basic discount rate is usually the cost of capital to the enterprise. For ranking the projects according to this method, the NPVs of various alternative projects are compared. Project with highest positive NPV or a project with highest NPV is given highest rank.

Accept-Reject Rule:

In the case of independent projects, if the present value of cash inflows of a project is higher than the present value of investment outlay of the project, it should be accepted. Otherwise, it should be rejected. In the case of mutually exclusive projects, a project with highest NPV should be accepted.

Advantages:

- It is simple to understand
- Where a company has several mutually exclusive projects in hand, this method helps the management to choose the most profitable one.

Disadvantages:

- It does not take into consideration the magnitude of the investment outlay and net cash benefits together.

Internal Rate of Return (IRR):

This rate tries to find the earnings rate which equates the present value of the streams of earnings to the investment outlay. IRR is defined as the rate of return which discounts all the future cash inflows to exactly equal the outlay.

Accept-Reject Rule:

The project with IRR higher than the cut-off rate will be accepted. Otherwise, it will be rejected. The management will be indifferent if the IRR = cut-off rate.

Advantages:

- It is useful and has many positive points
- It recognizes the time value of money
- It helps the management in selecting the most profitable project

Disadvantages:

- It is complicated to calculate by trial and error method
- It assumes that the funds received at the end of each year can be invested at the same rate of return.
- It does not provide weight age of the volume of funds committed in the project.

- Under certain conditions it becomes very difficult to take any decisions like – under conditions of irregular cash flows, IRR may give 2 or more answers.

Profitability Index (PI):

It is a ratio of the present value of the net cash benefits to the present value of the net cash outlay. The higher the PI, the greater the return. Any project with a PI higher than ONE is acceptable since benefits exceed outlay. Projects with PI less than ONE are rejected.

Advantages:

- It places the present value of each investment project on a relative basis so that projects of different sizes of capital outlays can be compared.

Discounted Payback Period (DPBP):

The discounted payback period is the no of periods taken in recovering the investment outlay on the present value basis? Discounted payback period will always be higher than simple payback period for a project because its calculation is based on the discounted cash flows. It differs from the simple pay period in that it takes into account the time value of money. But still, it does not account for post pay back profitability of the project.

Factors influencing capital expenditure decisions:

1. Availability of funds

Generally, capital expenditure projects require large funds. A project, however profitable, may not be taken for want of funds. So, projects with a lesser profitability may be sometimes preferred due to lesser pay back period for want of liquidity.

2. Urgency

Sometimes an investment is to be made on the grounds of urgency for the firm's survival or to avoid heavy losses. In such circumstances, proper evaluation of proposal cannot be made through profitability test. E.g. break down of machinery, fire accident etc.

3. Legal Compulsion

When statutory compulsion arises, investment has to be made in a project though it may not be profitable one. For example, waste disposal plants have to be installed to satisfy environmental laws.

4. Degree of uncertainty

Profitability is directly related to risk. Normally, higher the profits, greater is the risk or uncertainty. Sometimes, a project with lower profitability may be selected due to constant flow of income as compared to another project with an irregular and uncertain flow of income.

5. Intangible factors:

Sometimes, a capital expenditure has to be made due to certain emotional and intangible factors such as safety and welfare of workers, prestigious project, social welfare, goodwill of the firm etc., though such investments are not profitable.

6. Obsolescence:

If obsolete plant and machinery exist in a firm, their replacement becomes necessary.

7. Research and Development

It is necessary for the long term survival of the business to invest in research and development projects though it may not loom to be a profitable investment.

8. Competitors' activities

When competitors perform certain activities, they may compel a firm to undertake similar activities to withstand competition.

9. Future Earnings

A project may not be profitable today when compared to another one, but it may promise better future earnings. In such cases, it may be preferred to increase earnings.

Capital budgeting process / Phases of capital budgeting:

The entire Capital budgeting process can be divided into following steps:

- 1) **Identification of potential investment opportunities:** the process of Capital budgeting begins with identifying potential investment opportunities. An individual or a planning committee is responsible for developing estimates of future sales, which form the basis for setting future production targets. Based on such information, estimates of required investment can be made. For imaginative identification of investment ideas it is helpful to
 - i) monitor external environment (PEST analysis) regularly to scout investment opportunities
 - ii) formulate a well defined corporate strategy based on a thorough analysis of strength weakness opportunities and threats
 - iii) share corporate strategy and perspective with persons who are involved in process of Capital budgeting
 - iv) motivate employees to make suggestions
- 2) **Assembling of investment proposals:** various investment proposals identified by departments of a company are submitted in a standardized capital investment proposal form. These proposals are routed through various persons in order to evaluate the capital investment decision from different angles. Projects can be classified as expansion replacement new product diversification or welfare projects.
- 3) **Decision Making:** the projects then undergo a preliminary screening to obtain those which merit further consideration. Different executives are vested with the authority to approve investment proposals to certain limits. For e.g. consider a manufacturing concern. The plant superintendent can approve investment outlays up to Rs 20,00,000. Any investment above the mentioned level needs the approval of the board of directors. Such a decision-making process breaks up the investment approval process. Different appraisal criteria are used for the project selection such as payback, NPV etc.
- 4) **Preparation of capital budget and appropriations:** Projects involving smaller outlays, decided at the lower levels of the management are covered by a blanket appropriation. While those involving large cash outlays are included in the budget after getting necessary approvals. A proper appropriation of expenses ensures adequacy of resources during implementation of the capital expenditure decisions.
- 5) **Implementation:** translating an investment decision into a concrete project is a time-consuming process. Delay in implementation can lead to substantial cost and time overruns. To ensure proper implementation, the following points must be kept in mind:

- Adequate formulation of projects- this involves, conducting preliminary studies and a comprehensive and detailed formulation. It brings to light any difficulties likely to be faced in future. Therefore adequate formulation is necessary to ensure right implementation
- Use of the principle of responsibility accounting- assigning specific responsibilities to project managers for completing the project within the defined time frame and cost limit is helpful for the correct implementation of projects
- Use of network techniques- using techniques like CPM, PERT, help easy implementation and monitoring of projects

6) **Performance review:** post completion audit is used as a feedback device. It compares the actual performance with the projected performance. Based on the review corrective steps can be taken. It is useful in following ways

- i) it throws light on how realistic were the assumptions underlying the project
- ii) it provides a documented log of experience that is highly valuable for decision making
- iii) it helps in uncovering judgment biases
- iv) It includes a desired caution among project investors.

Capital budgeting decisions/Project classification:

a) **Accept- reject decision-** this is the fundamental decision in capital budgeting. If the project is accepted then the firm invests in it, or else rejects it. In general all projects which yield returns higher than the required rate of return (cost of capital in most cases) are accepted and the rest are rejected. By these criteria all independent projects that satisfy the minimum investment criteria are accepted. An independent Project is a project whose cash flows are not affected by the accept- reject decision for the projects and the selection of one is not dependent on any other project

b) **Mutually exclusive projects:** are set of projects from which at the most one will be accepted. They are set of projects which are to accomplish the same task. The acceptance of one excludes the acceptance of other projects. For e.g. deciding between a capital intensive or labour intensive machine. Thus when choosing between mutually exclusive projects more than one project may satisfy the Capital Budgeting criterion. However only one i.e. the best project can be accepted.

c) **Capital rationing:** Capital rationing is a situation where a constraint or budget is placed on the total size of capital expenditures during a particular period. Often firms draw up their capital budget under the assumption that the availability of financial resources is limited. Capital rationing refers to a situation where a company cannot take all acceptable projects it has identified because of shortage of capital. Under this situation a decision maker is compelled to reject some of the viable projects because of shortage of funds.

Factors leading to capital rationing:

External factors: capital rationing may arise due to external factor such as imperfection of capital market or deficiencies in the market information, which may result in the unavailability of capital. Generally the market itself or the government will not supply unlimited amount of investment capital to company, even though the company has identified investment

opportunities which would be able to produce the required return. Because of these imperfections the firm may not necessarily get amount of capital funds to carry out all profitable projects.

Internal factors: capital rationing is also caused by internal factors which are as follows

- Reluctance to take resort to external finance in order to avoid further risk
- Reluctance to broaden the equity share base for fear of losing control
- Reluctance to accept some viable projects because of its inability to manage the firm in the scale of the operation.

d) **Replacement decisions:** in case of Replacement decisions the implications are different.

Developing cash flows for new projects or expansion projects is relatively straightforward. In such cases the initial investment, operating cash inflows and terminal cash inflow are the after tax cash flows associated with proposed projects. Estimating the relevant cash inflows for a replacement project is somewhat complicated because you have to determine the incremental cash inflow and outflow in relation to existing project. The three components of the cash flow stream of a replacement project are determined as follows

i) initial investment (cost of new assets+ net working capital required for the new asset)- (after tax salvage value realized from old asset+ net working capital required for the old asset)

ii) Operating cash inflows= operating cash inflow from new asset-cash inflow from old asset that has not been replaced.

iii) Terminal cash flow=(after tax salvage value of new asset+ recovery of net working capital associated with the new asset)-(after tax salvage value of old asset , it had not been replaced+ recovery of net working capital associated with the new asset)

Features of capital budgeting decisions:

1. Existence of potentially large anticipated benefits
2. Involves a relatively high degree of risk
3. Existence of a relatively long time period between the initial outlay and the anticipated returns
4. Generally they are irreversible without incurring loss
5. They involve large volume of capital
6. They are strategic investment decisions and involve blocking of funds for long term and inevitably affect the firm's future cost structure
7. They affect the profitability of a firm
8. They have a bearing on the competitive position of the firm mainly because of the fact that they relate to fixed assets, which in a sense are the earning assets of the firm
9. Cash flow and not profit are important in investment decisions

Types of Investment Decisions / Proposals

1. Mutually exclusive investment proposals
2. Independent investment proposals
3. Contingent investment proposals
4. Replacements

COST OF CAPITAL

A project's Cost of Capital is the minimum acceptable rate of return/required rate of return on funds committed to the project. It is a compensation for time and risk in the use of capital by the project. Since the investment projects may differ in risk, each one of them will have its own unique cost of capital. The Firm represents the aggregate of investment projects undertaken by it. Therefore, the firm's Cost of Capital will be the overall or average, required rate of return on the aggregate of the investment projects.

- In NPV method, the Cost of Capital is the discount rate used for evaluating the desirability of an investment project.
- In IRR method, Cost of Capital is the minimum required rate of return on an investment project. It is also known as the cut off or target rate or the hurdle rate.
- The Cost of Capital is the minimum required rate of return on the investment project. It is also known as the cut off or target rate or the hurdle rate.
- The Cost of Capital is the minimum required rate of return on the investment project that keeps the present wealth of shareholders unchanged.

Basic Assumptions of Cost of Capital

The Cost of Capital is a dynamic concept affected by a variety of economic and firm factors and assumes the following assumptions relating to risk and taxes:

- 1) Business Risk: It refers to the risk of the inability of the firm to cover its operating costs. This cost is assumed to be unchanged i.e. the firm's acceptance of a given project does not affect its ability to meet operating costs.
- 2) Financial Risk: It refers to the risk of the inability of the firm to cover the required financial obligations (interests, lease payments or preference dividend) – is assumed to be unchanged.
- 3) After Tax Costs: They are considered relevant. The Cost of Capital is measured on an after tax basis.
- 4) Capital Structure: The firm's financial structure is assumed to remain fixed.

Concepts of Cost of Capital

1. Explicit Cost And Implicit Cost

The explicit cost of any source of finance may be defined as the discount rate that equates the present value of the funds received by the firm net of underwriting costs, with the present value of expected cash outflows. These outflows may be interest payments, dividend, or repayment of principal.

The implicit cost is the rate of return on the best investment opportunity for the firm and its shareholders which will be foregone if the project presently under consideration by the firm is accepted.

2. Future Cost And Historical Cost

Future cost refers to the expected cost of funds to finance the proposed project whereas Historical cost refers to the cost already incurred for financing a particular project.

3. Specific Cost And Combined Cost

The Cost of each component of capital (i.e. equity shares/preference shares/debentures/loans/etc) is known as Specific Cost of Capital. The Composite/Combined

Cost of Capital is the total cost of capital from all sources (i.e. equity shares/preference shares/debentures/loans/etc).

4. Average Cost And Marginal Cost

The average cost of capital refers to the weighted average of the cost of each component of funds employed by the firm. The weights are in proportion of the share of each component of the share in the total capital structure. Marginal Cost of Capital is the weighted average cost of new funds raised by the firm.

Factors affecting the cost of capital of a firm

1. The capital structure of a firm
2. The demand supply forces existing in the market for debt or equity
3. The existing and potential growth prospects of the firm
4. The risk profile of the majority shareholders
5. The stage of the firm's lifecycle etc

CAPITAL RATIONING

Introduction:

Capital budgeting decisions involve huge outlay of funds. Funds available for projects may be limited. Therefore, a firm has to prioritize the projects on the basis of availability of funds and economic compulsion of the firm. It is not possible for a company to take up all the projects at a time. There is the need to rank them on the basis of strategic compulsion and funds availability. Since companies will have to choose one from among many competing investment proposal the need to develop criteria for Capital rationing cannot be ignored. The companies may have many profitable and viable proposals but cannot execute because of shortage of funds. Another constraint is that the firms may not be able to generate additional funds for the execution of all the projects. When a firm imposes constraints on the total size of firm's capital budget, it is requires Capital Rationing. When Capital is rationed there is a need to develop a method of selecting the projects that could be executed with the company's resources yet giving the highest possible net present value.

Meaning of Capital Rationing:

Because of the limited financial resources, firms may have to make a choice from among profitable investment opportunities. Capital rationing refers to a situation in which the firm is under a constraint of funds, limiting its capacity to take up and execute all the profitable projects. Such a situation may be due to external factors or due to the need to impose internal constraints, keeping in view of the need to exercise better financial control.

Why Capital Rationing?

Reasons for Capital Rationing:

Capital Rationing may be due to

- a. External factors
- b. internal constraints imposed by management.

External Capital Rationing: External Capital Rationing is due to the imperfections of capital markets Imperfection may be caused by:

- a. Deficiencies in market information
- b. Rigidities that hamper the force flow of Capital between firms.

When capital markets are not favourable to the company the firm cannot tap the capital market for executing new projects even though the projects have positive net present values. The following reasons attribute to the external capital rationing:

1. Inability of the firm to procure required funds from Capital market because the firm does not command the required investor's confidence.
2. National and international economic factors may make the market highly volatile and instable.
3. Inability of the firm to satisfy the regularity norms for issue of instruments for tapping the market for funds.
4. High Cost of issue of Securities I,e High floatation cost. Smaller firms smaller firms may have to incur high costs of issue of securities. This discourages small firms from tapping the capital markets for funds.

Internal Capital Rationing: Impositions of restrictions by a firm on the funds allocated for fresh investment is called internal capital rationing. This decision may be the result of a conservative policy pursued by a firm. Restriction may be imposed on divisional heads on the total amount that they can commit on new projects. Another internal restriction for Capital budgeting decision may be imposed by a firm based on the need to generate a minimum rate of return. Under this criterion only projects capable of generating the management's expectation on the rate of return will be cleared. Generally internal capital rationing is used by a firm as a means of financial control.

Problems & Solutions

1. An asset has an initial investment of Rs. 6,00,000 with an anticipated life of 4 years. The estimated net annual cash inflows will be Rs. 1,50,000 , Rs, 2,00,000, Rs. 3,00,000 and Rs, 2,00,000 during the year I, II, III and IV respectively. Calculate the internal rate of return. (May/June 2007)

Solution:

Years	CIF	DF@10%	Present Value	DF @15%	Present value
1	1,50,000	.909	1,36,350	.870	1,30,500
2	2,00,000	.826	1,65,200	.756	1,51,200

3	3,00,000	.751	2,25,300	.658	1,97,400
4	2,00,000	.683	1,36,600	.572	1,14,400
			6,63,450		5,93,500
Less: Initial investment			6,00,000		6,00,000
Net present value			63,450		(6,500)

$$IRR = MTR + \frac{NPV \text{ of the LTR}}{\text{Diff between the HTR and LTR}} \times \text{Diff in TR}$$

$$= 10\% + \frac{63,450}{69,950} \times 8\%$$

$$IRR = 10\% + 4.53 = 14.52$$

2. From the following information, rank the projects according to their desirability under (i) Pay back period, (ii) Accounting rate of returns and (iii) Net present Value Index method assuming the cost of capital is 10%. **(Nov/Dec 2007)**

Projects	Initial Investment (Rs.)	Annual CIF (Rs.)	Life in years
A	60,000	12,000	15
B	88,000	22,500	22
C	2,150	1,500	3
D	20,500	4,500	10
E	4,25,000	2,25,000	20

You may use the following table for calculation:

Period in Years	3	10	15	20	22
P.V of an annuity of Re.1 pa payable for 'n' years at 10%	2.5918	6.3213	7.7688	8.6466	8.8919

Solution:

Projects	CIF	DF@10%	GPV	Investment	NPV
A	12,000	7.7688	92,225	60,000	32,225
B	22,500	8.8919	2,00,067	88,000	1,12,067
C	1,500	2.5918	3,888	2,150	1,738
D	4,500	6.3213	28,445	20,500	7,495
E	2,25,000	8.6466	19,45,485	4,25,000	15,20,485

Pay back Period Method

PB = Initial Investment / ACIF

A = 60,000 / 12,000 = 5 years - V

B = 88,000 / 22,500 = 3.9 years - III

C = 2150 / 1,500 = 1.43 years - I

D = 20,500 / 4,500 = 4.55 years - IV

E = 4,25,000 / 2,25,000 = 1.88 years - II

ARR = (Average Annual Income / Original Investment) * 100

A = 12,000 / 60,000 * 100 = 20% V

B = 2,500 / 88,000 * 100 = 2.56% III

C = 1,500 / 2150 * 100 = 69.76% I

D = 4,500 / 20,500 * 100 = 21.95% IV

E = 2,25,500 / 4,25,00 * 100 = 52% II

PI = PV of cash inflows / PV of cash outlays

A = 92,225 / 60,000 = 1.53 - IV

B = 2,00,067 / 88,000 = 2.27

C = 3,888 / 2150 = 1.80 - III

D = 28,445 / 20,500 = 1.38 - V

E = 19,45,485 / 4,25,000 = 4.57

-II

3. The following is the Capital structure of M/S Kurukshetra Earning Works Ltd., (Nov/Dec 2007)

Source of Finance	Amount(Rs)	Projections
Equity share Capital (45,000 shares of Rs. 10 each fully paid)	4,50,000	45%
Retained Earnings	1,50,000	15%
Preference share capital	1,00,000	10%
Debentures	3,00,000	30%
	10,00,000	100%

The firm's after tax component costs of the various sources of funds are as follows:

Equity capital	14%
Preferences share capital	10%
Retained earnings	14%
Debt	4.5%

Calculate the weighted cost of capital

- By assigning the book-value as weight and
- Market values as weight. Assume the market price of equity shares is Rs. 20 per share.

Solution:

Composite cost of capital under BOOK VALUE approach

Sources	Amount (Rs.)	Proportion (weights)	CC	WCC
Equity Capital	4,50,000	.45	.14	0.063
Retained Earnings	1,50,000	.15	.14	0.021
Preference Capital	1,00,000	.10	.10	0.010
Debt capital	3,00,000	.30	.045	0.013
Total	10,00,000	1.00		0.107 (or) 10.7%

Composite Cost of capital under MARKET VALUE approach

No of Equity shares = $4,50,000 + 1,50,000 = 6,00,000 / 10 = 60,000$

*Market Value = $60,000 * 20 = 12,00,000$

Sources	Amount (Rs.)	Proportion (weights)	CC	WCC
Equity Capital	12,00,000	.75	.14	0.105
Preference Capital	1,00,000	.06	.10	0.006
Debt capital	3,00,000	.19	.045	0.008
Total	16,00,000	1.00		0.119 (or) 11.9%

4. Which Investment is a risky one from the following returns? (May/June 2008)

BHEL 12% 14% 16% 18% 20%

SBI 12% 15% 20% 16% 17%

RIL 15% 21% 23% 20% 16%

Solution:

Range = Highest return – Lowest return

BHEL 20%-12% =8%

SBI 20%-12% =8%

RIL 23%-15% =8%

Standard deviation (Risk)

BHEL = 3.16

SBI = 2.91

RIL = 3.39

Therefore RIL is Risky. Since, it has more volatile. ($\sigma=3.39$)

5. A company's debentures with face value of Rs.100 bear an 8 percent coupon rate. Debentures of this type currently yield 10 percent.

i) What is the market price of debentures of the company?

Given

$$\text{Face Value of Debenture} = \text{Rs.100}$$

$$\text{Coupon Rate} = 8\%$$

$$\text{Current Yield} = 10\%$$

Market price of debentures of the company

$$\begin{aligned} \text{Market Price} &= \text{INT} / (1 + kg) + D / (1+ke) \\ &= 8 / (1+10) + 100 / (1+10) \\ &= 7.27 \times 90.90 \\ &= \text{Rs.98.17} \end{aligned}$$

ii) What would happen to the market price of the debentures if interest rises to 16 percent?

$$\begin{aligned} \text{Market Price} &= \text{INT} / (1+ke) + D/(1+ke) \\ &= 16 / (1+10) + 100 / (1+10) \\ &= 14.55 + 90.90 \\ &= \text{Rs.105.46} \end{aligned}$$

iii) What would be the market price of the debentures if it is assumed that debentures were having a maturity period of 4 years from now (bases on situation (i)? (Nov / Dec 2008)

$$\begin{aligned} \text{Market Price} &= \text{INT} / (1+ke) + \text{INT} / (1+ke)^2 + \text{INT} / (1+ke)^3 + \text{INT} / (1+ke)^4 + D / (1+ke)^4 \\ &= 8 / (1+10) + 8 / (1+10)^2 + 8 / (1+10)^3 \\ &= +8 / (1+10)^4 + 100 / (1+10)^4 \\ &= 7.27 \times 6.60 + 6.01 + 5.46 + 68.30 \\ &= \text{Rs.93.65} \end{aligned}$$

iv) Would you pay Rs.90 to purchase debentures specified in situation (iii) above. Explain.

If market price is Rs.93.65, it is advisable to Pay Rs.90 to purchase debentures

6. A project has the following pattern of cash flows

Year	0	1	2	3	4	5
Cash flow (Rs)	-40,00,000	15,00,000	8,00,000	7,50,000	-8,00,000	35,23,000

i). Calculate IRR of the project (ii) with $i=8\%$ calculate NPV of the project. (May/June 2009)

Year	Cash Flow	Discounting Factor	@12%	@ 13%
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		@8%		P.V.	Discounted Cash flows	P.V.	DCF
		Preset Value	Discounted Cash Flows				
1	1500000	0.926	1389000	0.893	1339500	0.885	1327500
2	800000	0.857	685600	0.797	637600	0.783	626400
3	750000	0.794	595500	0.712	534000	0.693	519750
4	-800000	0.735	588000	0.636	508800	0.613	490400
5	3523000	0.681	2399163	0.568	2001064	0.543	1912989
Total Discounted cash inflows			4481263		4003364		3896239

Cash Outflow = Rs.4000000

Total Discounted Cash Inflows

at 12% = Rs.4003364

at 13% = Rs.3896239

It clearly shows that IRR lies between 12% and 13%. In order to find out the exact IRR, the following formula can be applied.

$$\begin{aligned}
 \text{IRR} &= 12\% + \frac{4003364 - 4000000}{4003364 - 3896239} \times (13\% - 12\%) \\
 &= 12\% + \frac{3364}{107125} \times 1\% \\
 &= 12\% + 0.03\% \\
 \text{IRR} &= 12.03\%
 \end{aligned}$$

ii) **Calculation of NPV at 8%**

Total Discounted Cash inflows at 8% = 4481263

(-) Cash Outflows = 4000000

Net Present Value (NPV) = 481263

7. Enumerate is considering investing in a new products. The following annual information is available. (May/June 2009)

- * Cash sales Rs.25 Lakhs
- * Credit sales Rs.32 lakhs
- * Manufacturing cost of sales, excludes depreciation and includes Rs.75,000 of allocated fixed cost Rs.16 lakhs
- * Selling a administrative expenses (includes Rs.5,00,000 of allocated costs) Rs.7 Lakhs
- * Loss of contribution on other products Rs.2 Lakhs

Credit sales are collected after a year with a bad debt loss of 2%. The capital equipment costs Rs.30 lakhs and is depreciated at 33 1/3% per annum, written down value method. Working capital investment of Rs.20 lakhs is required.

A long-term debt carrying 14% interest rate will increase by Rs.20 lakhs and a short term bank borrowing carrying 18% rate will increase by Rs.10 lakhs. The corporate tax rate is 50%

Compute the cash flows for four years.

Ans : Particulars	Years			
	1	2	3	4

Cash Sales	2500000	2500000	2500000	2500000
Credit Sales	3200000	3200000	3200000	3200000
Total Revenue (1)	5700000	5700000	5700000	5700000
Manufacturing Cost	1600000	1600000	1600000	1600000
Selling and Adm. Exp.	700000	700000	700000	700000
Bad Debt. Losses	64000	64000	64000	64000
Depreciation	1000000	666667	444444	296296
Total Payments (2)	3364000	3030667	2808444	26602963
Cash Flow (1-2)	2336000	2669333	2891556	3039704
(-) Tax @ 50%	1168000	1334667	1445778	1519852
Net Cash Flow	1168000	1334666	1445778	1519852

Calculation of Depreciation under WDB Method

Cost of Equipment = 3000000

(-) Dep. @ 33 1/3% = 1000000

~~2000000~~

(-) Dep. @ 33 1/3% = 666667

~~1333333~~

(-) Dep. @ 33 1/3% = 444444

~~888889~~

(-) Dep. @ 33 1/3% = 296296

~~592593~~

(-) Dep. @ 33 1/3% = 197531

8. ABC Limited is proposing to invest in a project requiring a capital outlay of Rs.50,000. Cost for annual income after depreciation but before tax is as follows :

Year	1	2	3	4	5
Rs.	20,000	20,000	16,000	16,000	8,000

Depreciation may be taken at 20% on original cost and taxation at 50% of net income. You are required to evaluate the project according to each of the following method. (May/June 2009)

- Pay-back method.
- Rate of return on original investment method
- Rate of return on average investment method
- Discount cash flow method taking cost of capital at 10%
- Excess present value index.

Solution:

Calculation of Adjusted cash inflows:

Particulars	1	2	3	4	5
Cash inflow after Depreciation (-)	20000	20000	16000	16000	8000
) Tax @ 50%	10000	10000	8000	8000	4000
Cash Inflow after Tax (+)	10000	10000	8000	8000	4000
Depreciation	10000	10000	10000	10000	10000
Cash Inflow after tax before dep.	20000	20000	18000	18000	14000

i) Calculation of Pay Back Method:

Year	Cash Inflows	Cumulative Cash Inflows
1	20000	20000
2	20000	40000
3	18000	58000

4	18000	76000
5	14000	90000

Cash Out flow = Rs.50000

$$\text{Pay - Back Period} = 2 + \frac{10000}{18000} = 2 \text{ Years}$$

ii) Rate of Return on Original Investment Method :
Average Annual Cash Inflows

$$\frac{\text{Original Investments}}{(10000+10000+8000+8000+4000) / 5} \times 100$$

$$= \frac{50000}{8000} \times 100$$

$$= 6250\%$$

iii)

$$\frac{\text{Average Annual Cash Inflows}}{\text{Average Investments}} \times 100$$

$$= \frac{8000}{50000 / 2} \times 100$$

$$= 32\%$$

iv) Discount cash flow method taking cost of capital at 10%

Year	Cost of Capital @ 10%	Cash Inflows	Discounted Cash Inflows
1	0.909	20000	18180
2	0.826	20000	16520
3	0.751	18000	13518
4	0.683	18000	12294
5	0.621	14000	8694
			69206

Total Discounted Cash Inflows = 69206
(-) Cash Outflows = 50000
Net Present Value = 19206

v)

$$\frac{\text{Discounted Cash Inflows}}{\text{Cash Outflows}} \times 100$$

$$= \frac{69206}{50000} \times 100$$

$$= 138.41\%$$

9. From the following structure of a company, calculate the overall cost of capital, using Book value weights, Market value weights

Equity shares (Rs.10/Share) 45,000 90,000

Retain earnings 15,000

Preference share capital 10,000 10,000

Debentures 30,000 30,000

The after tax cost of different sources of finance is as follows :

Equity : 14% Preference Share capital : 10%

Retained Earnings : 13% Debentures : 5%, (May/June 2009)

Solution:

i) Calculation of Overall Cost of Capital using Book Value Weights :

Sources	Book Value	Weights	After-Tax Cost	Weighted Cost of Capital
Equal shares	45000	0.45	14%	6.30
Retired Earnings	15000	0.15	13%	1.95
Pref. Shares	10000	0.10	10%	1.00
Debentures	30000	0.30	5%	1.50
Total	100000			10.75

ii) Calculation of Overall cost of Capital using Market Value Weights:

Sources	Book Value	Weights	After-Tax Cost	Weighted Cost of Capital
Equal shares	90000	0.692	14	9.69
Pref. Shares	10000	0.077	10	0.77
Debentures	30000	0.231	5	1.16
Total	130000			11.62

10. A limited company is considering investment in a project requiring a capital of Rs. 2, 00,000.

Forecast for annual income after depreciation but before tax is as follows: (MAY/JUNE 2012)

Year	Rs.
1	1,00,000
2	1,00,000
3	80,000
4	80,000
5	40,000

Depreciation may be taken as 20% on original cost and taxation at 40% of the net income. You are required to evaluate the project according to each of the following methods.

(i) pay-back method

(ii) Accounting rate of return on original investment

(iii) Discounted cash flow method taking cost of capital as 10%

(iv) Internal rate of return method

Solution:

(i) pay-back method

Statement of cash inflow

Year	Profit	Tax at 50%	Profit after tax	Depreciation	Cash inflows	Cumulative cash

						inflows
1	1,00,000	50,000	50,000	40,000	90,000	90,000
2	1,00,000	50,000	50,000	40,000	90,000	1,80,000
3	80,000	40,000	40,000	40,000	80,000	2,60,000
4	80,000	40,000	40,000	40,000	80,000	3,40,000
5	40,000	20,000	20,000	40,000	60,000	4,00,000

Payback period = Rs. 1, 80,000 is recovered in 2 years

The balance of Rs. 20,000 = $20,000/80,000 \times 1 = 0.25$ years

The Payback period is 2.25 yrs. or 2 years and 3 months.

(ii) Accounting rate of return on original investment

Year	Net Profit after tax but before depreciation
1	50,000
2	50,000
3	40,000
4	40,000
5	20,000
Total Profits	Rs. 2,00,000

Average annual profits = Rs. 2, 00,000/ 5 = Rs.40, 000

Rate of return = $40,000/2, 00,000 \times 100 = 20\%$

(iii) Discounted cash flow method taking cost of capital as 10%

Year	Cash inflows	Discount factor @ 10 %	Present value in Rs.
1	90,000	0.909	81,810
2	90,000	0.826	74,340
3	80,000	0.751	60,080
4	80,000	0.683	54,640
5	60,000	0.621	37,260
Present value of cash inflows			3,08,130
Less : Initial Investment			2,00,000
Net Present Value			1, 08, 130

(iv) Internal rate of return method

= $40,000/1, 00,000 \times 100 = 40\%$

11. From the following capital structure of a company, calculate the overall cost of capital, using

(i) Book weight

(ii) Market value weights

Sources	Book value (Rs.)	Market value (Rs.)
Equity share capital (Rs. 10 shares)	45,000	90,000
Retained earnings	15,000	-----
Preference share capital	10,000	10,000
Debentures	30,000	30,000

The after tax cost of different sources of finance is as follows:

Equity share capital – 14%; retained earnings – 13%

Preference share capital – 10%; debentures – 5% . (May/June 2012)

Solution:

i).Calculation of Overall Cost of Capital using Book Value Weights:

Sources	Book Value	Weights	After-Tax	Weighted
---------	------------	---------	-----------	----------

			Cost	Cost of Capital
Equal shares	45000	0.45	14%	6.30
Retired Earnings	15000	0.15	13%	1.95
Pref. Shares		0.10	10%	1.00
Debentures	10000	0.30	5%	1.50
	30000			
Total	100000			10.75

ii) .Calculation of Overall cost of Capital using Market Value Weights:

Sources	Book Value	Weights	After-Tax Cost	Weighted Cost of Capital
Equal shares	90000	0.692	14	9.69
Pref. Shares	10000	0.077	10	0.77
Debentures	30000	0.231	5	1.16
Total	130000			11.62

12. A Company is considering two mutually exclusive projects. Both require an initial cash outlay of Rs. 10,000 each and have a life of five years. The company's required rate of return is 10% and pays tax at 50 % rate. The projects will be depreciated on a straight line basis. The before taxes cash flow expected to be generated by the projects are as follows:

	Before tax cash flow (Rs.)				
Project 1	2	3	4	5	
A	4000	4000	4000	4000	4000
B	6000	3000	2000	5000	5000

Calculate for each project (i) the payback (ii) The ARR (iii) the NPV (iv) IRR. Which project should be accepted and why? (May/June 2013)

Answer - Project A

Particulars	Amount
Cash flows before tax	4,000
Less : Tax @ 50%	2,000
EAT	2,000
Add: Depreciation	2,000
Earnings after tax but before depreciation	4,000

(i) Pay back period = Investment / AACIF = 10,000/4,000 = 2.5 years

(ii) ARR = AACIF (After dep & tax) / Average investment
= 2,000/ 5,000 = 40%

(iii) NPV = CIF * Df @10% (.909+.826+.751+.683+.621)
= 4000*3.79 = 15,160 – 10,000 = 5,160

(iv) IRR = F = I/C = 10,000/4,000 = 2.5 = 28%

Answer - Project B

Depreciation = 10,000/5= 2,000

CIF (Before tax)	Tax 50%	CIF (after tax)	CIF (Before after dep, tax)	Df @11%	Present Value
6,000	3,000	3,000	5,000	.909	4545
3,000	1,500	1,500	3,500	.826	2891

2,000	1,000	1,000	3,000	.751	2253
5,000	2,500	2,500	4,500	.683	3073
5,000	2,500	2,500	4,500	.621	2794
		10,500			15,556
Gross Present Value					15,556
Initial Investment					10,000
NPV					5,556

(i) Pay back period = 3 year + (311/3073) = 3.10 years

(ii) ARR = Average CIF ((After dep & tax) / Average investment
= Average CIF = 10,500/5 = 2,100
= 2,100/5000 = 42%

(iii) IRR

CIF (Before dep, after tax)	Df @10%	Present Value	Df @20%	Present Value	Df @30%	Present Value
5000	0.909	4545	0.833	4165	0.769	3845
3500	0.826	2891	0.694	2429	0.592	2072
3000	0.751	2253	0.579	1737	0.455	1365
4500	0.683	3073	0.482	2169	0.350	1575
4500	0.621	2794	0.402	1809	0.269	1210
Gross present Value		15556		12309		8702
Initial Investment		10000		10000		10000
NPV		5556		2309		(1298)

IRR = 10% + 5556 / (5556 - (-1298)) * 20% = 26.21%

	Project A	Project B	Decision
Payback Period	2.5 years	3.1 years	Project A accepted
NPV	5160	5556	Project B accepted
ARR	40%	42%	Project B accepted
IRR	28%	26.21%	Project A accepted

13. Sagar Industries is planning to introduce a new product with a projected life of 8 years. The project, to be set up in a backward region, qualifies for one time (as its starting) tax free subsidiary from the government of Rs. 20 lakhs. Initial equipment cost will be needed at the beginning of the third year. At the end of 8 years, the original equipment will have no resale value, but the supplementary equipment can be sold for Rs.1 lakh. A working capital of Rs. 15 lakhs will be needed. The sales volume over the eight years period have been forecasted as follows:

Years	Units
1	80,000
2	1,20,000
3-5	3,00,000
6-8	2,00,000

A sale price of Rs. 100 per unit is expected and Variable expenses will amount to 40% of sales revenue. Fixed cash operating costs will amount to Rs. 16 lakhs per year. In addition, an extensive advertising campaign will be implemented, requiring annual outlays as follows:

Years Rs. (in lakhs)

1	30
2	15
3-5	10
6-8	4

The company is subject to 50% tax rate and considers 12% to be an appropriate after –Tax cost of capital for this project. The company follows the straight line method of depreciation. Should the project be accepted? Assume that the company has enough income from its existing products. (May/June 2008)

Solution:

Cash outflow

Initial equipment cost 140,00,000

Less Tax free subsidiary 20,00,000

120,00,000

Add: Working expenses 15,00,000

135,00,000

Additional equipment in the III year

10,00,000 *0.797 = 7,79,000

Total Cash outflow = 1,35,00,000 +7,79,000

= 142,97,000

Cash inflow:(Rs. in Thousands)	1	2	3-5	6-8
Sales A	80,000	1,20,000	3,00,000	2,00,000
Variable cost	32,000	48,000	1,20,000	80,000
Fixed cost	16,000	16,000	16,000	16,000
Advertisement	30,000	15,000	10,000	4,000
Depreciation	17,500	17,500	19,000	19,000
Total Cost B	95,500	96,500	1,65,000	1,19,000
Profit (A-B)	(15,500)	23,500	1,35,500	81,000
Tax	-	4,000	67,500	40,500
PAT	(15,500)	19,500	67,500	40,500
Cash inflow (PAT+ Dep.)	2,000	37,000	86,500	59,500

Rs. in Thousands

Years	CIF	DF @12%	Present Value
1	2,000	.893	1,786
2	37,000	.797	29,489
3	86,500	.712	61,588
4	86,500	.636	55,014
5	86,500	.567	49,046
6	59,500	.507	30,167
7	59,500	.452	26,894
8	59,500	.404	24,038
WC	15,000	.404	6,060
Scrap	1,000	.404	404
Gross present value			284,28,500

Less: Initial Investment	142,97,000
Net Present Value	141,51,500

NPV is positive; therefore the project will be accepted